

International Islamic University Chittagong
Center for General Education (CGED)
Semester End Examination, Spring-2024

Course Code: URED-3604(URED-3201 for LLB)
Course Title: Life and Teachings of the Prophet (SA'AS)

Full Marks: 50

Time: 2 Hours 30 Minutes

(Answer all questions. The right side columns contain marks, CLOs, and Bloom's taxonomy domain for each question):

#	Questions	Marks	CLOs	Bloom's taxonomy domain
1	Explain the background, importance and lessons of the Hijrah of Prophet (SAAS) from Makkah to Madinah.	10	3	Create
2	a) Who were the Jewish tribes that lived in Madinah? Why and how were they expelled from Madinah? Or, b) What are the main clauses of the Charter of Madinah? How did Prophet (SAAS) establish a society based on peaceful co-existence in Madinah? Explain.	10	3	Understand & Create
3	Analyze the background of the battle of Ahzab assessing the reasons of victory of the Muslims in the battle.	10	3	Analyze & Evaluate
4	"The treaty of Hudaibiah is a clear victory"- explain elaborately.	10	3	Create
5	Evaluate the Farewell Address of Prophet Muhammad (SAAS) as a document of human rights.	10	3	Evaluate

International Islamic University Chittagong

Center for General Education (CGED)

Semester End Examination: Spring 2024

Course Code: GEHE-3601

Program: Undergraduate

Course Title: History of Emergence

Bangladesh

Full Marks: 50

Time: 2 hours and 30 minutes.

Instructions:

i. All Questions are Compulsory.

ii. Figures in the right margin indicate full marks.

iii. Course Learning Outcome (CLO) and Bloom's levels are mentioned in additional columns.

Bloom's Levels of the Questions.					
Letter of Symbol	R	U	App	An	E
Meaning	Remember	Understand	Apply	Analyze	Evaluate

	Text of the Questions	Marks	Bloom's Level	CLO
1	Evaluate the trend and nature of the students' movement of 1962 against General Ayub Khan and estimate its impact on the subsequent Bangali nationalism movement.	10	E	CLO2
2	Assess the historical context of the Mass Upsurge of 1969 highlighting the significant events during the movement.	10	An	CLO3
3	Review the spontaneous resistance and sector-based strategic battle during the nine-month-long Liberation War of Bangladesh. Or Explain the factors and disparities that disintegrated Pakistan and led to the emergence of independent Bangladesh in 1971.	10	E	CLO3
4	Evaluate the role of India in the Liberation War of Bangladesh.	10	E	CLO3
5	Write short notes on any two: a) United Front of 1954 b) Operation Searchlight c) Mujibnagar Government.	5 5	E	CLO3

International Islamic University Chittagong
Department of Computer Science and Engineering
B. Sc. in CSE Final Examination, Spring 2024
Course Code: CSE-3525 Course Title: Data Communication
Total Marks: 50
Time: 2 hours 30 minutes

(i) The figures in the right-hand margin indicate full marks (ii) Course Outcomes and Bloom's Levels are mentioned in additional Columns

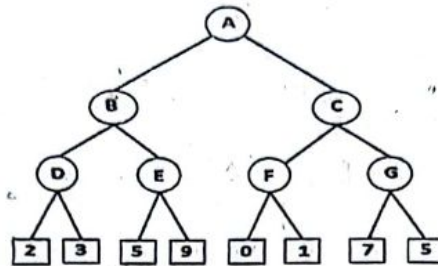
<u>Part-A</u>		M	C	DL
1. a)	A corporation possesses a satellite channel with a bandwidth of 10 MHz. The corporation intends to establish 40 distinct and independent channels, each capable of transmitting a minimum of 10 Mbps. Design a suitable modulation technique for this communication requirement.	5	3	A
1. b)	Four sources, each with a bit rate of 800 kbps, need to be combined using Time Division Multiplexing (TDM) with a byte interval and synchronizing bits. Answer the following questions regarding the multiplexing process: i. What is the size of a frame in bits? ii. What is the frame rate? iii. What is the duration of a frame? iv. What is the data rate? Or	5	3	A
1. b)	If there are 5 baseband analog signals, how many carriers are required when you are doing Frequency Division Multiplexing? Write differences between Time Division Multiplexing and Frequency Division Multiplexing.	5	3	A
2. a)	Wireless systems usually use analog transmission for transmitting the digital data. In this case environment, distance, bitrate etc. are important issues to be considered. Using data "11100011" draw the waveform for 4FSK, 4PSK techniques. Write your comments on each of these modulation techniques.	5	3	U
2. b)	Draw the constellation diagram for 8 QAM, 4 phases, two amplitudes and show the time domain plot for the given digital information for the signal 001110100010000101. Or	5	3	A
2. b)	Illustrate Binary BFSK and BPSK for the data 111110. Assume these bits travel in 1 second.	5	3	A
<u>Part-B</u>				
3. a)	Explain with proper diagrams what would happen if a frame is lost, an acknowledgement is lost and consecutive acknowledgements are lost for selective repeat ARQ flow control.	5	2	A
3. b)	Four 1 kbps connections are multiplexed together. A unit is 1 bit. Find (a) the duration of 1 bit before multiplexing, (b) the transmission rate of the link, (c) the duration of a frame, (d) duration of a time slot.	5	4	A
4. a)	If a flag is 01111100, then explain how would you do bit stuffing if this pattern occurs in the data. Or	5	3	A
4. a)	We have an available bandwidth of 200 KHz which spans from 300 KHz to 500 KHz. What are the carrier frequency and bit rate if we modulated our data by using Binary ASK with $d=1$?	5	3	A
4. b)	Suppose the sender has encoded the data word 1001101 using a Hamming code for error correction. However, the receiver receives the codeword 10010100101. Explain whether the receiver has received the correct codeword or not. If not, correct the error.	5	3	A
5. a)	Analyze Frequency Reuse in Cellular Networks with figure.	5	4	A
5. b)	Explain Basic Service Set and Extended Service Set in Wireless LAN with figure. Or	5	1	U
5. b)	Write short notes of the followings (Any Five, 5 x 1): i. Satellite Communication, ii. Mobile Communication, iii. Ethernet, iv. Wireless LAN v. Bluetooth, vi. SONET, vii. STS.	5	1	U

International Islamic University Chittagong
Department of Computer Science and Engineering
B. Sc. in CSE Final Examination, Spring 2024
Course Code: CSE 3635 Course Title: Artificial Intelligence
Total marks: 50
Time: 2 hours 30 minutes

The figures in the right hand margin indicate full marks.
 Course Outcomes and Bloom's Levels are mentioned in additional Column

Part A

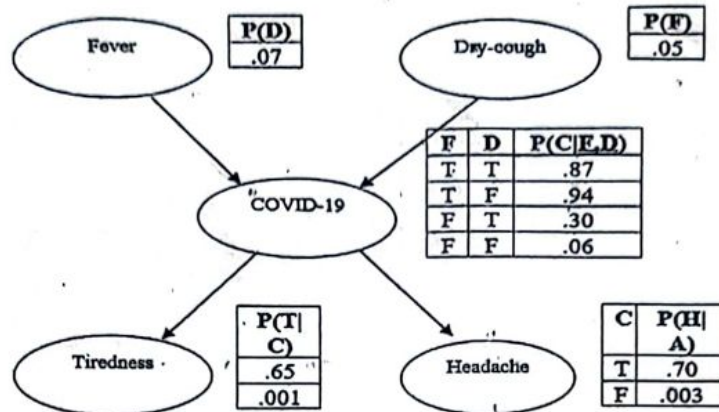
- | | | C | DL | Marks |
|---|---|---|----|-------|
| 1 | a) Consider the following game tree in which static scores are all from the first player's point of view: | 3 | U | 5 |



- | | | | | |
|---|--|---|----|-------|
| | i) Suppose the first player is the maximizing player. What move should be chosen?

ii) In the game tree, what nodes would not need to be examined using the alpha-beta pruning procedure? | 3 | | 5 |
| 1 | b) What is CSP? Do you consider the map coloring as an example of CSP? If so, identify the variables, domains, and constraints associated with this problem. | 3 | An | 5 |
| | OR | | | |
| 1 | b) Analysis between forward chaining and backward chaining. In what aspects forward chaining should be used? | 3 | An | 5 |
| 2 | a) In a knowledge-based system, the following rules are given:
Rule 1: A breeze in cell B22 implies that there is a pit in either cell P21, P23, P12, or P32.
Rule 2: There is no breeze in cell B22.
Using logical rules such as biconditional elimination, modus ponens, disjunction elimination, and other necessary logical rules, determine
$\neg (P21 \vee P23 \vee P12 \vee P32)$ | 3 | Ap | 5 |
| | OR | | | |
| 2 | a) Consider the following sentences.
i) All people who are graduating are happy
ii) All happy people smile
iii) Someone is graduating.
iv) Is someone smiling?
A) Convert the above sentences into FOL.
B) Convert the sentences (FOL) into clausal form
C) Prove (iv) by using resolution. | 3 | Ap | 2+2+1 |

2. b) The following Bayesian network, showing both the topology and the conditional probability tables (CPTs). In the CPTs, the letters C, F, D, T and H stand for COVID-19, Fever, Dry-Cough, Tiredness, and Headache respectively. The Independent conditional probability helps us to write in a simplified way the joint distribution $P(C, F, D, T, \text{ and } H)$.



- (i) Express the joint distribution $P(C, F, D, T, H)$ in terms of the conditional probabilities (and independencies) expressed in the Bayesian Network above.
- (ii) Probability of the event that the patient is tested positive for COVID-19 but the patient does not have symptoms of neither a Fever nor Dry-cough and both symptoms of tiredness and headaches have occurred.

Part B

- 3 a) Explain the Bayes Theorem. 4 U 3
- 3 b) About 2/3 of drivers use their cell phone while driving. Suppose that you are 5 times more likely to get into an accident if you text and drive, and if you don't use your cell phone, you have a 1% chance of getting into an accident. What is the probability that someone was texting given that they got into an accident? 4 Ap 5

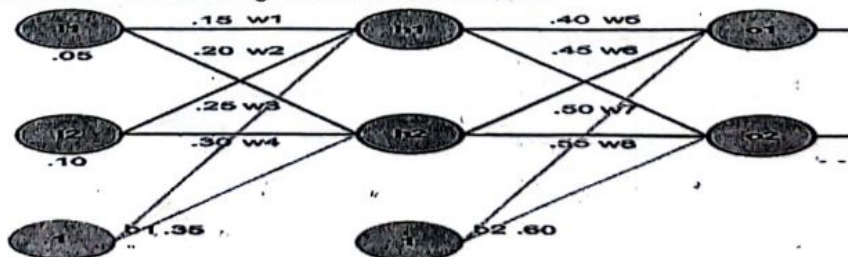
OR

- 3 b) In a medical scenario, it is known that patients with dengue fever(+d) develop a rash(+r) 80% of the time. The doctor also has the following information: the prior probability of a patient having dengue fever is 1/10000 and the probability that a patient has rash given that they do not have dengue fever is 1/100. Find the probability that a patient has a dengue given that they have rash. 4 Ap 5

- 3 c) What is NLP? Write down the steps through which NLP is conducted. 4 R 2

- 4 a) How do neural networks differ from conventional computers? Briefly explain. 4 U 2

- 4 b) Consider the following neural network model. 4 Ap 5

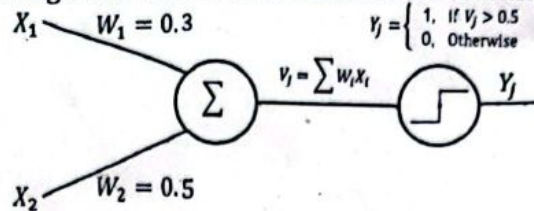


- i) Compute the value of O1 and O2 without transfer function.
- ii) Compute the value of O1 and O2 with sigmoid transfer function.

all error goes to next neuron.

OR

Consider the following neural network model with a learning rate of 0.2.



4 Ap 5

Demonstrate the learning process of the OR operation as shown below using the delta rule.

- c) Considering the target output $o_1=0.01$ and $o_2=0.99$ in the network depicted in question 4(b), calculate the error for each output neuron as well as the total error of the network using the squared error function. 4 U 3

- a) Write down the top down and bottom up parse tree for the sentence "Book the flight through Houston" using the following CFG. 4 Ap 4

Grammar	Lexicon
$S \rightarrow NP VP$	$Det \rightarrow the a that this$
$S \rightarrow Aux NP VP$	$Noun \rightarrow book flight meal money$
$S \rightarrow VP$	$Verb \rightarrow book include prefer$
$NP \rightarrow Pronoun$	$Pronoun \rightarrow I he she me$
$NP \rightarrow Proper-Noun$	$Proper-Noun \rightarrow Houston NWA$
$NP \rightarrow Det Nominal$	$Aux \rightarrow does$
$Nominal \rightarrow Noun$	$Prep \rightarrow from to on near through$
$Nominal \rightarrow Nominal Noun$	
$Nominal \rightarrow Nominal PP$	
$VP \rightarrow Verb$	
$VP \rightarrow Verb NP$	
$VP \rightarrow VP PP$	
$PP \rightarrow Prep NP$	

- b) Suppose the reputation of a university depends on it's academic excellence and infrastructure. These attributes are graded as poor, average, or good, with specific weights and degrees of belief assigned to each evaluation grade for assessment. 3 Ap 4

Evaluation Grade	Weight	Belief		
$H_1=\text{poor}, H_2=\text{average}, H_3=\text{good}$				
	W_i	$\beta_{1,i}$	$\beta_{2,i}$	$\beta_{3,i}$
Accuracy	0.35	0.4	0.5	0
Reputation	0.65	0.1	0.75	0.15

Calculate the remaining belief and probability mass of the following attributes using DS theory.

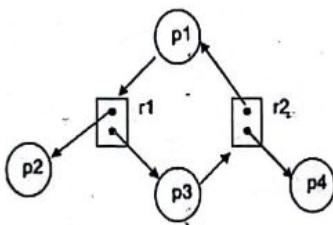
Evaluation Grade	Belief	Probability mass					
$H_1=\text{poor}, H_2=\text{average}, H_3=\text{good}$							
	β_H	$m_{1,i}$	m_2	$m_{3,i}$	$m_{H,i}$	$\sim m_{H,i}$	$\sim m_{H,i}$
Accuracy							
Reputation							

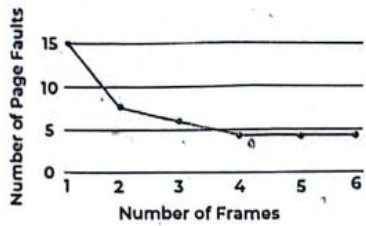
- c) What do you mean by uncertainty? What are the different sources of uncertainty. 3 U 2

International Islamic University Chittagong
Department of Computer Science & Engineering
B. Sc. in CSE Semester Final Examination, Spring 2024
Course Code: CSE 3631 Course Title: Operating Systems
Total marks: 50. Time: 2.5 hours

Answer all Five of the following questions. The figures in the right-hand margin indicate full marks.

Group - A

		M	C	D														
1.a)	A system has three processes (P0, P1, and P2) competing for three resource types (A, B, and C). The following information is available: Max Matrix (Max): This matrix specifies the maximum number of resources of each type each process may ever request. Allocation Matrix (Alloc): This matrix shows the current number of resources of each type allocated to each process Available Matrix (Avail): This matrix shows the number of available resources of each type in the system. <table border="1"> <thead> <tr> <th>Process</th> <th>Resources -A,B,C-Max</th> <th>Resources- AB,C- Alloc</th> <th>Resources- A,B,C-Available</th> </tr> </thead> <tbody> <tr> <td>P0</td> <td>7,5,3</td> <td>0,1,0</td> <td rowspan="3">3,2,1</td> </tr> <tr> <td>P1</td> <td>3,2,2</td> <td>2,0,0</td> </tr> <tr> <td>P2</td> <td>9,0,2</td> <td>1,0,0</td> </tr> </tbody> </table> 1. Is the system in a safe state? Explain your reasoning using the Banker's algorithm. 2. Suppose process P2 requests one additional unit of resource A (Request = [1, 0, 0]). Will granting this request lead to a safe state? Explain your answer using the Banker's algorithm.	Process	Resources -A,B,C-Max	Resources- AB,C- Alloc	Resources- A,B,C-Available	P0	7,5,3	0,1,0	3,2,1	P1	3,2,2	2,0,0	P2	9,0,2	1,0,0	8	2	
Process	Resources -A,B,C-Max	Resources- AB,C- Alloc	Resources- A,B,C-Available															
P0	7,5,3	0,1,0	3,2,1															
P1	3,2,2	2,0,0																
P2	9,0,2	1,0,0																
b)	Is there a deadlock in this scenario based on the Resource Allocation Graph? Explain your reasoning using the four necessary conditions for deadlock. (Mutual Exclusion, Hold and Wait, No Preemption, Circular Wait) 	2	2															
2.a)	Consider a swapping system in which memory consists of the following hole sizes in memory order: 10 MB, 4 MB, 20 MB, 18 MB, 7 MB, 9 MB, 12 MB, and 15 MB. Which hole is taken for successive segment requests of 12 MB, 10 MB and 9 MB for best fit? Compare the time complexity of best fit, first fit and worst fit. System Parameters:	3	3															
b)	This scenario analyzes the impact of page fault rate on the effective memory access time (EMAT) in a demand paging system. Main Memory Access Time (MMAT): 10 nanoseconds (ns) Disk Access Time (DAT): 100 milliseconds (ms) = 100,000,000 ns Page Hit Ratio (PHR): This value will be varied in the problem. Page Size: 4096 bytes	5	3															

	Calculate the Effective Memory Access Time (EMAT) for different page hit ratios (PHR) ranging from 0.9 (90% hits) to 0.1 (10% hits). Analyze the results.		
c)	Considering factors like page table size overhead and potential reduction in page faults, discuss the potential benefits and drawbacks of using a larger page size (8 KB) compared to a smaller one (2 KB).	2	3
	OR		
2.a)	Define contiguous memory allocation and explain its main advantages and disadvantages. Provide examples to illustrate external and internal fragmentation.	3	3
b)	Describe the role of the Translation Lookaside Buffer (TLB) in a paging system. How does it improve the efficiency of address translation?	4	3
c)	Consider a system with the following sequence of memory allocations and deallocations: Allocate 10 KB, Allocate 20 KB, Deallocate 10 KB, Allocate 5 KB, Allocate 15 KB. If the system initially has 50 KB of free memory, demonstrate how each memory allocation strategy (first-fit, best-fit, worst-fit) would handle these requests. Discuss the resulting fragmentation for each strategy.	3	3
	Group - B		
3.a)	Explain how the logical address is mapped to physical address with TLB.	3	3
b)	Consider the following page reference string: 1, 2, 3, 4, 2, 1, 5, 6, 2, 1, 2, 3, 7, 6, 3, 2, 1, 2, 3, 6. How many page faults would occur for the second chance replacement algorithms, assuming three frames? Remember that all frames are initially empty, so your first unique pages will cost one fault each. Compare this algorithm with FIFO page replacement algorithm considering page fault rate.	5	1
c)	The following chart represents the number of page fault vs number of frames. Analysis this chart with key findings. 	2	2
4.a)	If the logical memory space is 232 and page size is 4KB, how many entries is in the page table? Is it memory efficient to keep the Page table in this case? If not, write a solution to manage the page table efficiently?	5	4
b)	A system uses the working set model to guide page replacement decisions. A process has the following reference string: 1, 2, 3, 2, 1, 4, 3, 2, 1, 5, 2, 1 Assuming the working set window size (W) is 3, identify the pages that would be in memory at each reference using the working set model (consider a Least Recently Used - LRU replacement policy within the working set window).	5	4
5 a)	What are some best practices for securing a network against threats?	3	5
b)	How can program threats impact a computer system or network?	2	4
c)	A university computer system has three domains: Domain 1: Administrators, Domain 2: Professors, Domain 3: Students There are four objects in the system: Object 1: Student Grades File, Object 2: Course Materials Directory, Object 3: System Configuration File, Object 4: University Website The current access matrix is incomplete. You are tasked to define the access rights for each domain to ensure a secure system while allowing users to perform their tasks.	5	3

	Access Rights: Read (R), Write (W), Execute (X), None (-) Problem: Define the access rights for each domain (row) to each object (column) in the access matrix. Consider the following: Administrators should have full control over the system. Professors need to read course materials and student grades related to their courses. They should not modify student grades or the system configuration. Students need to read their own grades and access course materials relevant to their enrollment. They should not have access to other students' grades or the system configuration. The university website should be readable by everyone.																						
	OR																						
5 a)	Explain the concept of "breach of availability" in computer security. How can unauthorized destruction of data and denial of service (DOS) attacks impact the availability of a system or service? Provide examples of scenarios where these security violations might occur and discuss strategies for maintaining system availability despite potential threats.	3	3																				
b)	Suppose Alice wants to send another encrypted message to Bob. Here are Alice's new keys: - Alice's public key (e, n): (5, 91) - Alice's private key (d, n): (29, 91) Alice wants to send the message "M" to Bob, represented as a numerical value: $M = 12$. Using Alice's public and private keys mentioned above, calculate the encrypted value C of the message $M = 12$ following the RSA encryption process. Then, calculate the decrypted message "M" using Alice's private key. Provide the numerical values for C and M after the encryption and decryption steps.	3	3																				
c)	Consider an access matrix for a system with three subjects (S1, S2, S3) and four objects (O1, O2, O3, O4). Each entry in the matrix represents the access rights for a subject to an object, where 0 means no access, 1 means read access, and 2 means write access. The access matrix is as follows: <table border="1" style="margin-top: 10px;"> <thead> <tr> <th></th><th>O1</th><th>O2</th><th>O3</th><th>O4</th></tr> </thead> <tbody> <tr> <td>S1</td><td>1</td><td>2</td><td>0</td><td>1</td></tr> <tr> <td>S2</td><td>0</td><td>1</td><td>1</td><td>0</td></tr> <tr> <td>S3</td><td>2</td><td>0</td><td>0</td><td>1</td></tr> </tbody> </table> Calculate the total number of access rights granted in the access matrix.		O1	O2	O3	O4	S1	1	2	0	1	S2	0	1	1	0	S3	2	0	0	1	4	3
	O1	O2	O3	O4																			
S1	1	2	0	1																			
S2	0	1	1	0																			
S3	2	0	0	1																			

Bismillahir Rahmanir Rahim
International Islamic University Chittagong

Department of Computer Science & Engineering

B. Sc. in CSE, Final Examination, Spring 2024

Course Code: CSE-3637

Course Title: Software Engineering

Total marks: 50

Time: 2.5 hours

[Answer all the questions. Use separate answer scripts for Group A and Group B. Figures on the right margin indicate marks.]

[Group A]

CO Marks

1. Read the passage and answer questions 1(a-c)

Lets consider, You are managing the development of a new hospital management system (HMS) that is designed to streamline operations, improve patient care, and ensure compliance with healthcare regulations. The system will include modules for patient registration, appointment scheduling, billing, medical records and reporting.

As the project manager, you need to identify and categorize the stakeholders involved in the project to ensure that their needs and concerns are addressed throughout the development process.

- | | | |
|---|-----|---|
| 1(a) Identify the primary, secondary, and tertiary stakeholders for the hospital management system (HMS). | CO2 | 3 |
| 1(b) Explain the roles and interests of each category of stakeholders in the project. | CO2 | 4 |
| 1(c) Describe how you would engage each group of stakeholders to ensure the project's success. | CO2 | 3 |

- | | | |
|---|-----|---|
| 2(a) What is Software Design? Why Software Design is not Software Analysis? | CO3 | 3 |
| 2(b) What are the key principles of modularity in software design? | CO2 | 4 |
| 2(c) Write the differences between Coupling and Cohesion. | CO1 | 3 |

OR

- | | | |
|---|-----|---|
| 2(a) Briefly describe about socio-technical issue for software engineering. | CO3 | 5 |
| 2(b) What is JSP? Draw and explain JSP architecture | CO2 | 5 |

[Group B]

3. Read the passage and answer questions 3(a) & 3(b).

In a software development company, a team of skilled testers was responsible for ensuring the products' quality. They utilized various testing methods, such as black-box testing and white-box testing.

One day, the team was assigned a new project involving the creation of an online learning platform. As the project progressed, the testers were tasked with thoroughly testing the application before its release to identify any bugs or errors.

Now, consider that you are a member of the testing department. Using your knowledge of black box and white box testing, please answer the following scenario-based question.

You decide to employ both black box and white box testing techniques for the online learning platform as part of your testing strategy. The black box testing methodology entails testing an application from an external perspective, without knowing its internal structure or source code. In contrast, white box testing requires an examination of the application's internal workings, including the code, algorithms, and data structures.

- | | | | |
|------|--|-----|---|
| 3(a) | What specific scenarios or features would you focus on testing using black box techniques, and what scenarios or areas would you prioritize for white box testing? | CO4 | 6 |
| 3(b) | How would your choice of testing techniques contribute to ensuring the overall quality and functionality of the social networking application? | CO4 | 4 |
| 4(a) | Explain the software maintenance types of the following scenario:
(i) Software code optimization
(ii) Removing irrelevant software features
(iii) Introducing new cloud storage | CO4 | 3 |
| 4(b) | "Good software engineering and understanding how software is to be used by the customer are two crucial factors that affect software maintenance costs"- justify your answer. | CO1 | 3 |
| 4(c) | Elaborate on the process of software re-engineering and discuss the benefits of software re-engineering. | CO2 | 4 |

- | | | | |
|------|---|-----|---|
| 5(a) | Describe briefly one of the empirical factor models of software cost estimation techniques. | CO1 | 3 |
| 5(b) | Discuss the role of Henry and Kafura's Metric, Card and Glass's Systems Complexity metric. | CO2 | 3 |
| 5(c) | Suppose the fan out of a module is 5, the cumulative number of variables passed to and from the module is 7, the complexity of the module is 45. Use Card and Glass's Systems Complexity method in order to find out the structural complexity of the module. | CO3 | 4 |

OR

Read the passage and answer questions (1 - 3):

You are the project manager of a software development company that has been operating at CMM Level 2 (Repeatable) for the past two years. Your company has been successful in maintaining its project schedules and budget, and there is a set process for project management and quality assurance. However, the company has faced challenges with process variability, inconsistent project performance, and limited process improvement initiatives.

Your CEO wants to move the company to CMM Level 3 (Defined) within the next year to achieve more consistent project outcomes and to enhance the company's competitive edge. To do this, you need to implement standardized and documented processes across the organization.

- | | | | |
|------|--|-----|---|
| 5(a) | What key steps should you take to transition your company from CMM Level 2 to CMM Level 3? | CO2 | 3 |
| 5(b) | What challenges might you encounter during this transition, and how would you address them? | CO2 | 4 |
| 5(c) | How your company will reach CMM Level 3 benefit in terms of project performance and customer satisfaction? | CO2 | 3 |

Good luck

International Islamic University Chittagong
Department of Computer Science and Engineering
B. Sc. Engineering in CSE
Final Exam, Spring 2024

Course Code: ECON 3501
Time: 2 hours 30 minutes

Course Title: Principle of Economics
Full Marks: 50

- (i) Answer all the questions. The figures in the right-hand margin indicate full marks.
(ii) Course Outcomes (COs) and Bloom's Levels are mentioned in additional Columns.

Course Outcomes (COs) of the Questions	
CO1	Demonstrate knowledge of the fundamental concepts and theories of micro and macro-economics.
CO2	Identify the key indicators of economic growth and how to use the indicators for national and international comparisons.
CO3	Identify and explain the key macroeconomic policies.

Bloom's Levels of the Questions						
Letter Symbols	R	U	Ap	An	E	C
Meaning	Remember	Understand	Apply	Analyze	Evaluate	Create

Part A																																																	
1.	a)	Define market and market structure. Explain and compare the different types of market structure.	CO1	An	5																																												
1.	b)	The table below shows the market demand schedule and the cost structure. <table><tr><th>Quantity</th><th>Price per</th><th>Fixed cost (FC)</th><th>Variable cost (VC)</th></tr><tr><td>20</td><td>\$18</td><td>\$200</td><td>40</td></tr><tr><td>35</td><td>\$18</td><td>\$200</td><td>70</td></tr><tr><td>55</td><td>\$18</td><td>\$200</td><td>110</td></tr><tr><td>75</td><td>\$18</td><td>\$200</td><td>150</td></tr><tr><td>105</td><td>\$18</td><td>\$200</td><td>210</td></tr><tr><td>130</td><td>\$18</td><td>\$200</td><td>260</td></tr><tr><td>160</td><td>\$18</td><td>\$200</td><td>320</td></tr><tr><td>190</td><td>\$18</td><td>\$200</td><td>380</td></tr><tr><td>220</td><td>\$18</td><td>\$200</td><td>440</td></tr><tr><td>250</td><td>\$18</td><td>\$200</td><td>500</td></tr></table>	Quantity	Price per	Fixed cost (FC)	Variable cost (VC)	20	\$18	\$200	40	35	\$18	\$200	70	55	\$18	\$200	110	75	\$18	\$200	150	105	\$18	\$200	210	130	\$18	\$200	260	160	\$18	\$200	320	190	\$18	\$200	380	220	\$18	\$200	440	250	\$18	\$200	500	CO1	E	5
Quantity	Price per	Fixed cost (FC)	Variable cost (VC)																																														
20	\$18	\$200	40																																														
35	\$18	\$200	70																																														
55	\$18	\$200	110																																														
75	\$18	\$200	150																																														
105	\$18	\$200	210																																														
130	\$18	\$200	260																																														
160	\$18	\$200	320																																														
190	\$18	\$200	380																																														
220	\$18	\$200	440																																														
250	\$18	\$200	500																																														
Measure Total cost (TC), Average total cost (ATC) and Marginal cost (MC) for each row.																																																	
2.	a)	Suppose, GDP = \$5700 Net Factor payment from abroad = \$18 Capital consumption allowance = \$625 Indirect taxes = \$475 Social security contribution = \$530 Govt. and business transfers to person = \$770 Dividends = \$135 Personal Tax & Non-Tax payment = \$620 Required: Calculate GNP, NNP, National Income, Personal Income and Personal Disposable income	CO2	E	5																																												

2.	b)	Actual meaning of inflation is increase in money supply faster than the growth in real national income- analyze this statement with the reasons behind the inflation.	CO2	An	5
		OR			
2.	a)	Define the Value-added approach in measuring GDP with an example.	CO2	E	5
2.	b)	Generally nominal GDP always larger than real GDP. Why this happen? Can you explain?	CO2	An	5
		Part B			
		[Answer the questions from the followings]			
3.	a)	Discuss monetary policy and fiscal policy. How fiscal policy plays an important role in a developing country like Bangladesh?	CO2	An	5
3.	b)	Briefly explain unemployment with its classification. Evaluate the measures that can be taken to reduce unemployment problem in a country like Bangladesh.	CO2	E	5
4.	a)	Discuss the importance of planning in the economic development of Bangladesh.	CO1	Un	5
4.	b)	Not only the availability but also ability to utilize the natural resources is important in economic growth. – explain the statement in the case of Bangladesh.	CO1	E	5
5.	a)	Evaluate the relationship between Technological progress and economic growth in Bangladesh.	CO3	E	5
5.	b)	Explain the difference between growth and development? /	CO3	An	5
		OR			
5.	a)	Why some countries are rich and others are poor? Can you evaluate the matter in the context of Bangladesh and Singapore?	CO3	E	5
5.	b)	Political government alone is not responsible for development, there are some other partners. Who are those development partners? Discuss briefly.	CO3	An	5

International Islamic University Chittagong
Morality Development Program (MDP)
Semester Final Examination, Spring-2024
Faculty of Science and Engineering, Semester- 6th

Course Title: Islamization of Discipline

Course Code: MDP-3606

Time: 2:30 Hours

Total Marks: 50

=====

Answer the following questions; Figures in the right hand margin indicate full marks

1. a) What do you think about rational use of Information Technology in the light of Islam? 5
b) Highlight the ethical issues of computer use from the perspective of Islam. 5
2. a) Explain the physical and spiritual benefits of Salat and Fasting (Sawom) 6
b) What is autophagy? How autophagy works in the human body. 4
3. Write a brief summary of 4 Muslim Scientists mentioning their contributions to their respective fields 10
4. a) Define the term "disaster from Islamic perspective" and classify disasters with examples 3
b) What are the negative impacts of having sex outside of marriage, homosexuality, pornography and short dress on society? 5
c) Mention two examples of natural disasters described in the glorious Qur'an 2
- Or**
4. a) What is Miracle (Mojeza)? Mention four different miracles by four different messengers by mentioning Quranic verses. 5
b) The Qur'an itself is a miracle "- explain this statement with your logical argument 5
5. a) Establish the relationship between education and psychology 5
b) What factors are involved in teachers' responsibility to increase students' attention and interest in teaching and research? 5
- Or**
5. a) Mention the contribution of Bangabandhu Sheikh Mujibur Rahman in laying the foundations of religious liberalism in Bangladesh. 5
b) What significant role did Bangabandhu Sheikh Mujibur Rahman play in establishing Islamic consciousness and values? 5